

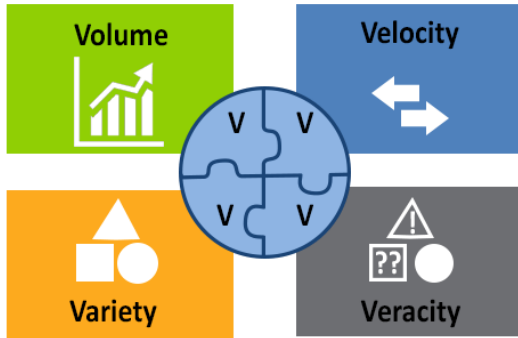
The background of the slide is a dark blue field filled with a complex, interconnected network of thin, light blue lines. These lines form a web-like structure with numerous nodes and clusters. Some nodes are highlighted with small, bright blue dots. The overall effect is a sense of digital connectivity and data flow.

Introduction to Big Data

COURSE INTRODUCTION

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Main topics in this course



Definitions, characteristics, and case studies, etc.



Hadoop DFS and MapReduce



Spark modules: Spark SQL, MLlib, GraphFrames, etc.



NoSQL databases and MongoDB



Stream processing

Weekly schedule

Week	Lecture topics
1	The Fundamentals of Big Data - Basic concepts and Characteristics - Frameworks and Technologies
2	Hadoop Fundamentals: Basic concepts and Ecosystem Hadoop Distributed Filesystem: HDFS daemons I/O operations
3	Hadoop MapReduce: MapReduce daemons Practice with Apache Hadoop
4	Spark Fundamentals: Basic concepts and Ecosystem Spark APIs
5	Advanced Analytics with Spark Practice with Apache Spark

Weekly schedule

Week	Lecture topics
6	NoSQL databases MongoDB in conjunction with Spark
7	Stream processing: Basic concepts Spark Streaming
8	A roadmap to GenAI tools
9-11	Seminar

Assignments, Exams, and Grading

- Grading

Midterm		Final		Seminar	Labs
15%	10%	20%	10%	25%	30%

- Midterm exam topics

- Big Data fundamental, Apache Hadoop, Apache Spark

- Final exam topics

- Spark APIs, NoSQL databases, Spark Streaming, and projects presented in the seminar

- Paper-based part (Mid: 60 mins – Final: 100 mins)

- In-class and open-book with only 2 A4 sheets.

- Coding part (All: 70 mins)

- At-home and open-book. Frameworks prepared in advance.

Assignments, Exams, and Grading

- Grading

Midterm		Final		Seminar	Labs
15%	10%	20%	10%	25%	30%

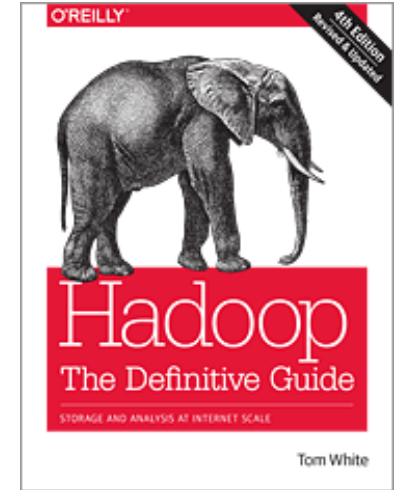
- Group assignments: 03-04 members/group recommended
- Seminar
 - Research on Big data topics and give oral presentation
- Lab work on Hadoop, Spark, and MongoDB
 - Install and explore the tools with simple case studies

Course regulations

- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- Students absent for the mid-term or final exam are considered unqualified for course completion.
- Students must accumulate at least 10% of course credits for lab work.
- Students will have several before-class assignments during the course. There will be a 1% penalty for each assignment missed.

List of reference books

- Tom White. 2015. *Hadoop: The Definitive Guide*. O'Reilly Media, Inc.



- Bill Chambers and Matei Zaharia, 2018. *Spark: The Definitive Guide: Big Data Processing Made Simple*. O'Reilly Media, Inc.

